



Temperature effect of photovoltaic cells: a review

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Abstract

The environmental problems caused by the traditional energy sources consumption and excessive carbon dioxide emissions are compressing the living space of mankind and restricting the development of economic society. Renewable energy represented by solar energy has gradually been moved to the forefront of energy development along with the strong support of national policies. Photovoltaic (PV) power generation is the main method in the utilization of solar energy, which uses solar cells (SCs) to directly convert solar energy into power through the PV effect. However, the application and development of SCs are still facing several difficulties, such as high cost, relatively low efficiency, and greater influence from external conditions. Among them, the temperature effect of SCs is related to their power generation efficiency, which is an important factor that needs to be considered in the development of SCs. The temperature effect of SCs will affect the intrinsic properties of SC materials and the parameters that characterize SC performance. This will ultimately affect its power generation efficiency. This work reviews previous studies on temperature effects in SCs. The influence of temperature effect on various parameters characterizing the performance of SCs is discussed, and its mechanism and the latest research progress are shown. It also introduces in detail various methods to deal with the temperature effect of SCs, and analyzes other factors that affect the performance of SCs. This review will help researchers in the design and development of SCs.

Keywords Solar cell · Photoelectric conversion · Temperature effect · Photovoltaic

Nomenclature

c-Si	Crystalline silicon
CB	Conduction band
CIGS	Copper indium gallium selenide
CdTe	Cadmium telluride
CEs	Counter electrodes
CIGSe	Cu(In,Ga)Se ₂
CPV	Concentrating photovoltaics
CELIV	Carrier extraction by linearly increasing voltage
DITC	Dark injection transient current
DC	Direct current

DSSCs	Dye-sensitized solar cells
DMOO	Dynamic multi-objective optimization
ETL	Electron transport layer
FF	Fill factor
HS	Heat sink
I _{sc}	Short-circuit current
J _{sc}	Current density
LID	Light-induced degradation
m-Si	Monocrystalline silicon
MJSC	Multi-junction solar cell
MPP	Maximum power point
MPPT	Maximum power point tracking
NFs	Nanofluids
OSCs	Organic solar cells
P3HT	3-Hexylthiophene
PCBM	[6,6]-Phenyl-C61-butyric acidmethyl ester
PSFPV	Partially submerged floating photovoltaic system
PV	Photovoltaic
P _{max}	Maximum power
PSCs	Perovskite solar cells
PVK	Perovskite
PCE	Power conversion efficiency

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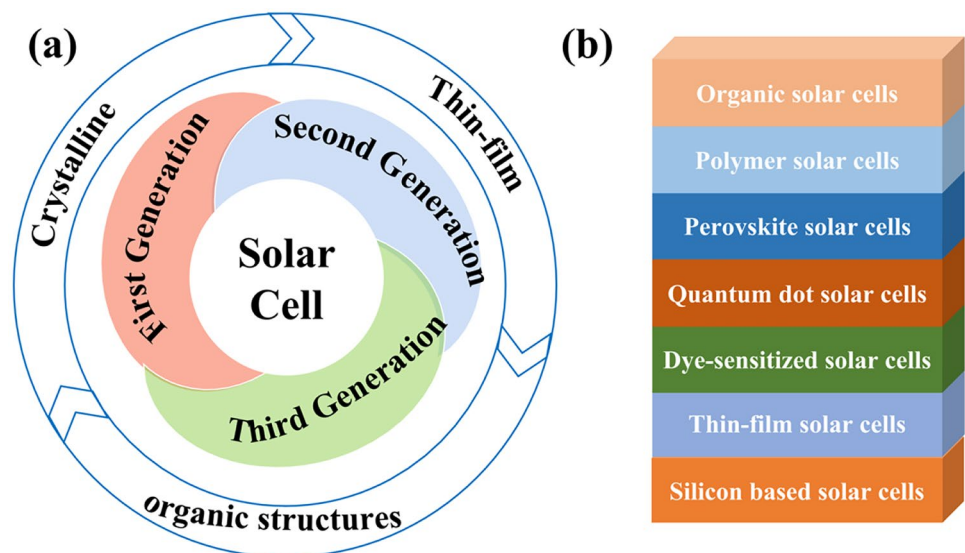
PC	Photocurrent
PL	Photoluminescence
PEG	Polyethylene glycol
PV/T	Photovoltaic/thermal
QDSCs	Quantum dot solar cells
QDs	Quantum dots
QE	Quantum efficiency
RC	Radiative cooling
SCLC	Space charge limited current
STC	Standard test conditions
SCs	Solar cells
TRPL	Time-resolved photoluminescence
TRPC	Time-resolved photocurrent
TCO	Transparent conductive oxide
Ti	Titanium
TEG	Thermoelectric generator
TE	Thermoelectric
UHCPV	Ultra-high concentration photovoltaic
VB	Valence band
V_{oc}	Open-circuit voltage
α -Si	Amorphous silicon
η	Photoelectric conversion efficiency

1 Introduction

Energy has always been an important factor leading to economic and social development. With the rapid consumption of traditional energy sources, greenhouse gases have continued to increase, making it difficult to contain the severe situation of global warming. Focusing on efficient energy conversion and adjusting the energy distribution structure have become the common choice of regulators and policy makers in many countries [1]. With the urgent need to deal

with climate change, both the “Copenhagen Agreement” and the “Paris Agreement” have pushed governments to set domestic emission reduction targets [2]. And countries all over the world are actively adding carbon peaking and carbon neutrality to their national development strategies. The new energy sources currently under active research mainly include solar energy, wind energy, nuclear energy, ocean energy, and biomass energy. And 70% of the energy consumed by the Earth comes from the sun [3]. The main utilization forms of solar energy are photothermal conversion [4, 5], photoelectric conversion [6, 7] and photochemical conversion [8, 9]. Photoelectric conversion is the process of directly converting solar radiation energy into power by using the PV effect of semiconductors [10]. At present, common photoelectric conversion devices are photoresistors [11], photodiode [12], SCs [3, 13, 14], and so on. Compared with other chemical cells such as lithium-sulfur cells [15–19] and zinc-ion cells [20, 21], SCs have unique natural advantages. Although current SCs are inferior to chemical batteries in terms of fabrication process and efficiency, they still have great potential. Since Bell Laboratories from America made practical monocrystalline silicon (m-Si) SCs for the first time in 1954 [22], SCs have experienced decades of development, as shown in Fig. 1a. The first generation of SCs is mainly based on crystalline silicon (c-Si) SCs [23]. But the cost of crystallization is too high to be commercialized, so the second generation of thin-film-based SCs came into being [3]. Along with the continuous advancement of science and technology, the third generation of SCs made with new materials and technologies has emerged. Such as organic structures of dye-sensitized solar cells (DSSCs), perovskite solar cells (PSCs), quantum dot solar cells (QDSCs), and so on [24]. And Fig. 1b lists the common SCs. Kant and Singh [25] have begun to study the fourth-generation

Fig. 1 **a** Development of SC technology, **b** several common types of SCs



SC technology that combines the low cost and flexibility of polymer films with the stability of organic nanostructures. This will make a significant contribution to the field of PV.

SCs are used in a wide variety of devices and are not limited to PV systems. For example, amorphous silicon (α -Si) SCs can be used in applications such as calculators, watches, and wristwatches [26]. PSCs can be combined with electrochemical energy storage systems such as supercapacitors and lithium-ion batteries [27]. Therefore, exploring the performance of SCs is crucial for its development. Li et al. [28] investigated the effects of ITO thickness and Na doping on the performance of Cu(In,Ga)Se_2 (CIGSe) thin-film SCs. Kumar et al. [29] studied the effect of ozone treatment on ZnO as electron transport layer (ETL) in organic solar cells (OSCs) by analyzing crystallinity, defect density of states, and charge carrier dynamics of transient absorption spectroscopy. Its purpose is to understand the role of ZnO in improving the interface between ETLs. Chatterji et al. [30] explored the temperature coefficients of silicon-based carrier-selective SCs. Meanwhile, to better understand and grasp the progress and blanks of SC research, researchers usually write some reviews so that people can better understand this field [31]. Rasal et al. [32] reviewed the stability of QDSSCs from the aspects of quantum dots (QDs), redox counterelectrolytes, counter electrodes (CEs), substrate types, and device packaging. And they made feasible suggestions for improving the stability of QDSSCs. Lee and Ebong [33] reviewed the technologies and challenges of α -Si, copper indium gallium selenide (CIGS), and cadmium telluride (CdTe). They described the reliability of these technologies as temperature-dependent, and the availability of materials as a decisive factor in determining which technologies can meet most of the future PV needs. However, most of the current review papers only review and summarize the effects on a particular class of cells and on a particular class of cells.

To our knowledge there is no article reviewing the performance of SCs based on temperature effects.

We know the PV modules are usually tested under standard conditions (i.e., standard test conditions (STC) are 1000 W/m^2 , AM1.5, 298.15 K), but the actual operating temperature is much higher and there are uncertainties [34]. As one of the core components of PV modules, solar panel performance is strongly influenced by its temperature. Moreover, different types of SCs respond differently to temperature. And the temperature coefficient of SCs is also affected by different factors. Compared to c-Si, thin-film SCs are less temperature-sensitive [34, 35]. The temperature coefficient of polymer SCs will be affected by the active layer [36, 37] and aging [38]. And the substrate temperature [39], the processing temperature [40], and the annealing temperature [41] will also affect the performance of SCs. In this work, the temperature effects of SCs will be reviewed from two main aspects, microscopic (Sect. 2) and macroscopic (Sect. 3). Other factors that affect the performance of SCs are also described (Sect. 4). And based on PV modules, we will further explore methods and measures to deal with temperature effects (Sect. 5). Finally, by reviewing previous studies we summarize the conclusions and give an outlook on the effect of temperature on SCs (Sect. 6).

2 Temperature effect on the micro-level

As shown in Fig. 2, SCs are defined as a component that directly converts photon energy into direct current (DC) through the principle of PV effect. Photons with energy exceeding the band gap of the cell material are absorbed, causing charge carriers to be excited, thereby generating current and voltage [42]. The effects of temperature on the microscopic parameters of SCs are interrelated and coupled.

Fig. 2 Schematic diagram of PV effect and PV conversion channel

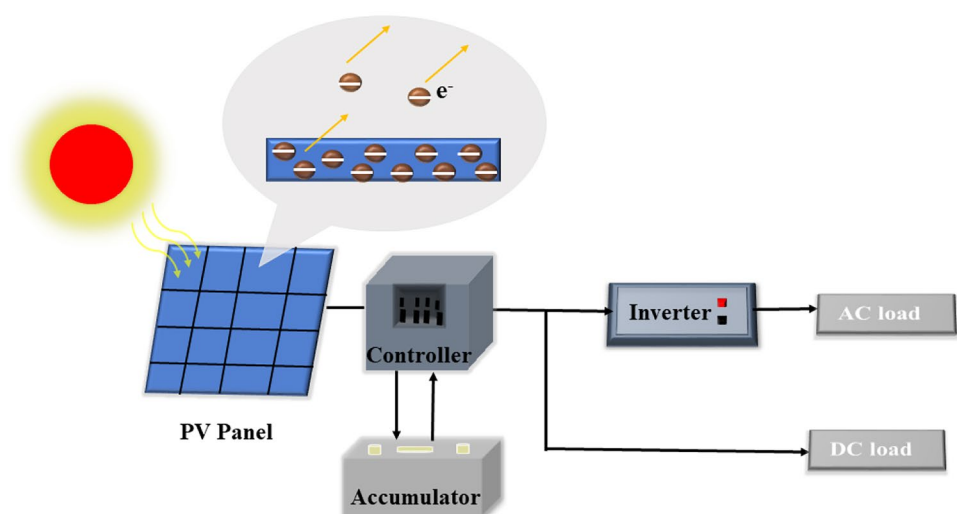


Table 1 Temperature effect on the micro-level

Microscopic level		Factors	Effect
Band gap Carrier	Mobility Lifetime	Temperature↑, Band gap↓,	The transport of carriers
		Temperature↑, mobility↓	Diffusion radius↓, η ↓
		Operating temperature↑, lifetime↓	Diffusion radius↓
		Annealing temperature↑, lifetime↑	Diffusion radius↑

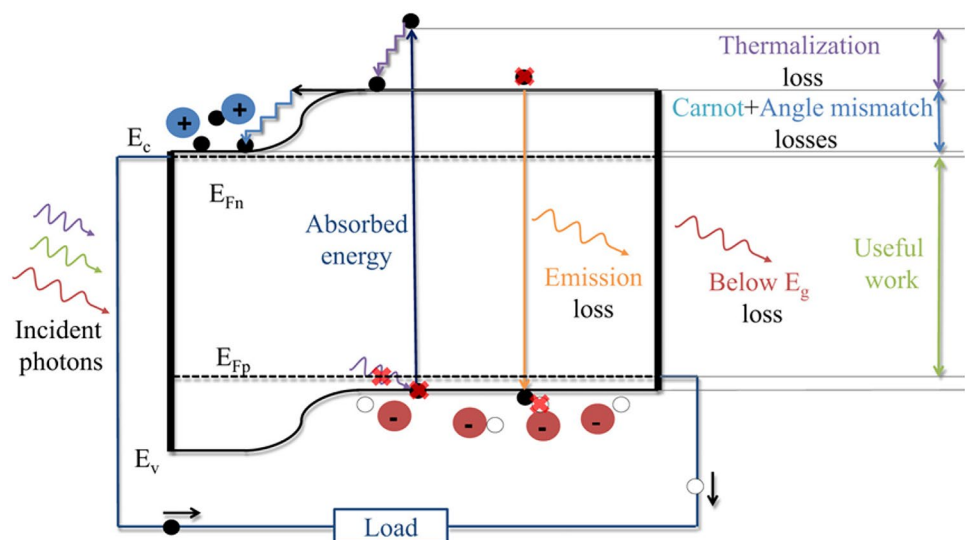
Table 1 summarizes the causes of temperature effects in SCs at the microscopic level and the resulting consequences. This section will clarify the relationship between these microscopic parameters under the influence of temperature effects from the aspects of semiconductor band gap, carrier mobility and carrier lifetime.

2.1 Temperature effect on the semiconductor band gap of SCs

Band gap, also known as energy gap and energy band gap, is one of the key factors affecting loss and SCs conversion efficiency. Only photons with energy higher than the forbidden band width can produce PV effect, which also determines the limit of the maximum wavelength that SCs can absorb for power generation [43]. And the higher the energy gap of the material, the smaller the temperature dependence of the SC parameter [44]. When semiconductors are disturbed by external energy such as temperature, pressure, photons, electric and magnetic fields, and the band gap changes. The band gap reflects the bond energy of the atoms in the cell. An increase in temperature changes the chemical bond and makes the electron transition from the valence band (VB) to

the conduction band (CB) [45]. Dupré et al. [46] proved the temperature dependence of the band gap and the influence of the incident spectrum on the temperature sensitivity of the short-circuit current (I_{sc}). And they showed the conversion loss mechanism of SCs in Fig. 3. They found that perovskite (PVK) semiconductor compounds (such as $\text{CH}_3\text{NH}_3\text{PbI}_{3-x}\text{Cl}_x$ and CsSnI_3) have band gap anomalous behavior, and this phenomenon causes the special temperature sensitivity of SCs in the radiation limit.

The higher the temperature, the lower the band gap of the SC, the cut-off wavelength will be longer, and the spectral range of its absorption will increase [45, 47]. However, the reduced band gap contributes to enhanced absorption of visible light, which leads to increased photons and hence photocurrents. Considering from the perspective of light, the increase in temperature is beneficial to PV power generation, because it will increase the free electron–hole pairs (i.e., carriers) generated by the PV effect in the cell to a certain extent [44]. However, excessively high temperature cannot increase the final output of the SC. Meanwhile, the transport of carriers inside the cell is also affected by the increase in temperature [48]. The transport of carriers inside the SC mainly depends on the drift effect of the built-in electric

Fig. 3 Conversion loss mechanism of SCs [46]

field and the photo-generated electric field and the diffusion effect of the carrier concentration. In other words, it is the mobility of carriers and the diffusion length of carriers. It is explained in detail below.

2.2 Temperature effect on carrier mobility in SCs

The carrier mobility is considered an important parameter of OSC materials because it affects charge extraction and recombination kinetics. The higher the mobility, the less time the charge carriers spend in the device and the lower the possibility of recombination [49]. Common methods for determining carrier mobility include steady-state space charge limited current (SCLC) [50], time of flight (TOF) [51], dark injection transient current (DITC) [52], and carrier extraction by linearly increasing voltage (CELIV) [53].

Ebenhoch et al. [54] studied the carrier mobility of organic PV materials PTB7 and PC₇₁BM and their influence on device performance through flight practical technology. In their work, the hole mobility of the mixture was reduced by a thousand times when the sample was cooled from room temperature to 77 K. The efficiency of the SC will be halved due to the loss of fill factor (FF) and I_{sc} . And the power conversion efficiency (PCE) as a function of temperature has a maximum value between 260 and 295 K. The research results showed that adjusting the mobility by changing the temperature is an effective method to determine the loss and optimization potential of the material. Sarkar et al. [55] conducted experimental studies on the carrier mobility of the ZnTTBPc-RGO heterojunction at different temperatures. In this study that the organic heterojunction of graphene oxide reduced by phthalocyanine was higher than the penetration threshold for PV applications. As can be seen in Fig. 4, the mobility of carriers in SCs increases with increasing temperature.

The mobility of carriers decreases with the increase of temperature, which leads to the deterioration of the output

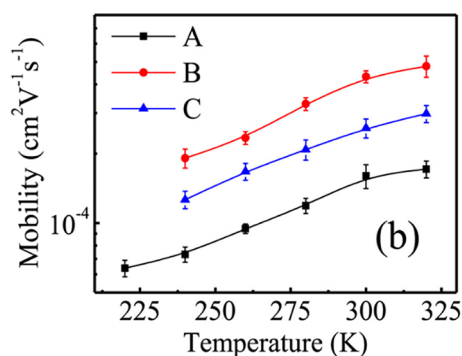


Fig. 4 Carrier mobility and error bars of ZnTTBPc-RGO heterojunction at different temperatures [55]

performance in the SC and the decrease of the photoelectric conversion efficiency (η). And the reduction of carrier mobility also reduces its diffusion radius, so that free carriers cannot be effectively transferred to the external circuit to form a current. In summary, the temperature increase has a negative effect on the drift effect of carriers inside the SCs.

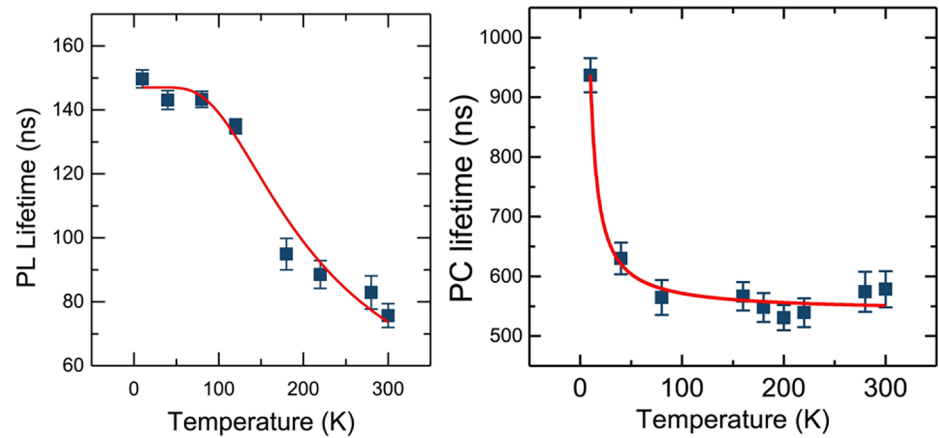
2.3 The temperature effect on the carrier lifetime of SCs

The increase in operating temperature not only affects the transfer process of carriers, but also reduces the lifetime of carriers. Carrier lifetime plays an important role in SCs, which is defined as the average time for excess minority carriers to return to equilibrium through various recombinations [56, 57]. When the SC is exposed to solar radiation to generate photo-generated carriers, the recombination center formed due to defects or impurities inside the cell will continuously capture free carriers. This phenomenon leads to a compound effect and a decreasing number of carriers. The stronger this recombination effect is, the less the number of carriers transported to the external circuit will eventually decrease its lifetime [58]. At present, the experimental techniques often used to detect the carrier lifetime in semiconductors and devices include time-resolved photoluminescence (TRPL) and time-resolved photocurrent (TRPC) [59–64]. Su and Xu [65] studied the effective lifetime of minority carriers in the time-resolved photocurrent (PC) and photoluminescence (PL) of doped semiconductors by modeling GaInP SC. One can see from Fig. 5, that the lifetime of PL decreases with the increase of temperature. When the temperature rises above 100 K, the PC lifetime of minority carriers tends to be flat. Hu et al. [66] studied the temperature and ligand-related carrier transport kinetics in PV PbS colloidal quantum dot films using the diffusion wave method. And they found that the carrier lifetime is monotonic dependent on temperature in the range of 100 to 300 K.

In SCs, the carrier lifetime is proportional to the diffusion length. The longer the minority carrier lifetime, the longer the diffusion length [67]. However, the annealing temperature will increase the lifetime of minority carriers [68]. O'Connor et al. [69] studied the effect of forming gas annealing on the inversion response and minority carrier generation lifetime of nandp-In_{0.53}Ga_{0.47}As MOS capacitors. In their research, it was observed that the minority carrier generation lifetime in In_{0.53}Ga_{0.47}As increased with the thickness of Al₂O₃ and increased after the forming gas annealing treatment.

Studies at the microscopic level have shown that the effects of temperature on the microscopic parameters of SCs are interrelated. Furthermore, the higher the band gap of the material, the smaller the temperature dependence of the SC parameter. And the forbidden band width indirectly affects

Fig. 5 The relationship between PL lifetime and temperature (left); the relationship between PC lifetime and temperature (right) [65]



the transport of carriers inside the SCs, thereby affecting the mobility and diffusion radius of carriers. And the carrier lifetime is directly proportional to the diffusion radius. This shows that too high a temperature rise can have a significant negative impact on the microscopic level of the SC.

3 Temperature effect on the macro-level

From a macro perspective, in the PV power generation process, SC needs to continuously receive radiation from sunlight. It must have the ability to withstand high-temperature conditions [70]. According to reports, the performance of PV modules is affected by the high temperature of solar panels (also called PV panels) [71]. And PV panels are also affected by the external environment, such as dust deposition [72], climate factor [73], etc. As the serviceable life decreases, the PV panels also experience aging, which also has a serious impact on the temperature effect of the PV panels or SCs [74]. Generally, electrical parameters such as open-circuit voltage (V_{oc}), FF, I_{sc} , current density (J_{sc}), η and maximum power (P_{max}) are used to express the temperature coefficient of SCs [75]. Table 2 shows the factors that cause temperature effects in SCs at the macroscopic level and the

resulting consequences. The following four parts will be used to illustrate the temperature effect of SCs in the macroscopic aspect of the external environment, non-uniform temperature, different types of SCs and their functional layers and the manufacturing process.

3.1 The external environment

Light intensity is one of the key factors affecting the performance of SCs. When continuously exposed to solar radiation, the temperature of SCs will rise significantly. Different bands of solar radiation can affect SCs in different ways. Many researchers have studied the effects of solar radiation and different spectra on the performance of SCs. To find the band when the PV panel effect and power conversion are optimal, Kazem and Miqdam [76] covered PV panels with filters of different colors. The findings show that covering the color filter reduces the performance of the PV panel, with the violet filter producing the highest current and voltage, due to the violet having the shortest wavelength and higher photon energy but lower efficiency. Whereas using a red filter produces lower current and voltage readings, red light produces more electricity than other spectral colors. Additionally, it has longer wavelengths, less photon energy,

Table 2 Temperature effect on the macro-level

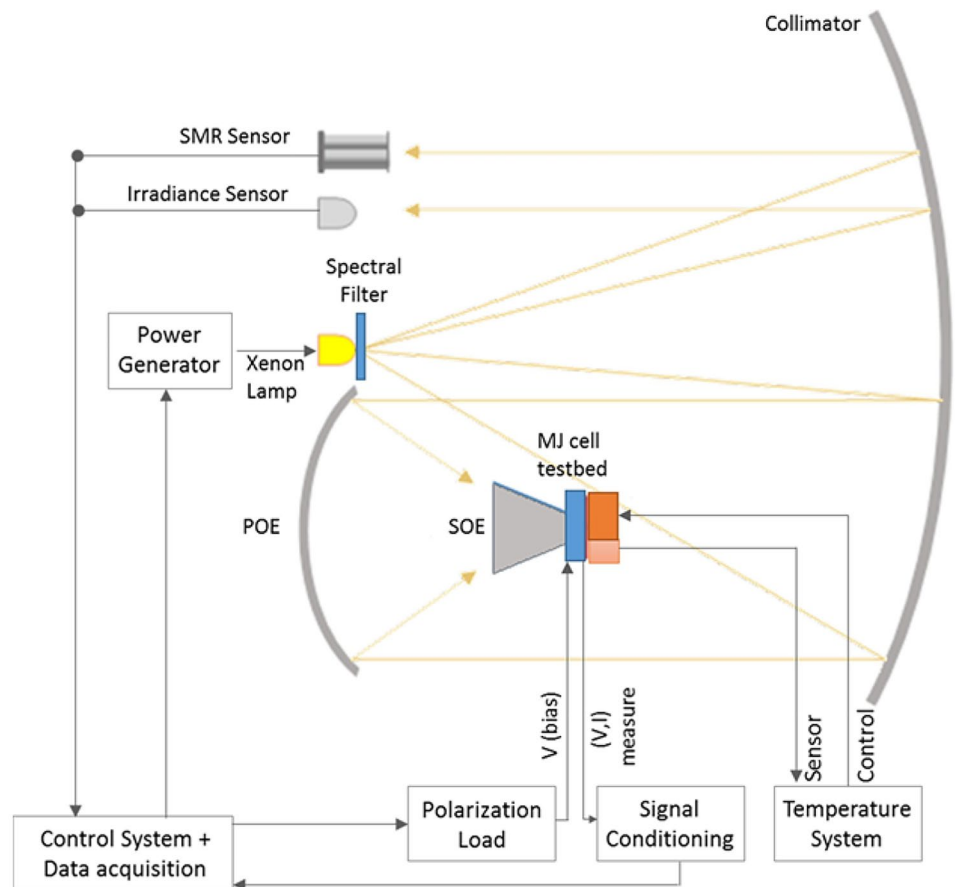
Macroscopic level		Factors	Effect
External environment	Light intensity	Solar radiation $\uparrow$, temperature $\uparrow$	$V_{oc}\uparrow$, $I_{sc}\downarrow$, $P_{max}\downarrow$, $\eta\downarrow$
	Dust deposition	Deposition $\uparrow$, temperature $\uparrow$	$\eta\downarrow$
	Environmental wind speed	Wind speed $\uparrow$, temperature $\downarrow$	$\eta\uparrow$
	Geographical location	Hot area, temperature effect $\uparrow$	η has a greater impact
Non-uniform temperature		Uneven lighting	
		Wind direction	
		Azimuth of wind	
		Inclination	
		Installation location of PV panels	
Azimuth $\uparrow$, temperature $\uparrow$			$\eta\downarrow$

and higher efficiency. Cotfas et al. [77] studied the effects of operating temperature (298.15–360.15 K) and irradiance (400–1000 W/m²) on the important parameters of m-Si-MSI, polycrystalline silicon-PSI, α -Si-ASI and multi-junction InGaP/InGaAs/Ge commercial SCs. The results showed that the V_{oc} of these four cells tended to increase with the increase of light intensity, while the I_{sc} decreased, resulting in a decrease in the P_{max} . Meanwhile, Basher and Kadhem [78] tested the relationship between solar radiation and current and voltage in SCs. The research results showed that as solar radiation increased, the temperature of the SC rose. The increase in solar radiation would increase the output current until the cell temperature was disturbed and caused it to drop. And the resulting voltage was not affected by the increase in current, but was more significantly affected by the increase in SC temperature. Concentrating photovoltaics (CPV) technology uses SCs and concentrating elements (such as Fresnel lenses) to convert solar radiation into power. Which will increase the amount of solar radiation incidence on the SC and cause the temperature of the cell surface and the surrounding environment to rise. In view of this, Nieto-Nieto et al. [79] proposed an experimental device to characterize the multi-junction solar cell (MJSC) of the ultra-high

concentration photovoltaic (UHCPV, irradiance level up to 1.5×10^6 W/m²) system (as shown in Fig. 6). Its purpose is to be able to use the device in the future to study the effects of temperature and spectral distribution deviations on SCs. Experimental investigations showed that this device is feasible. It is also expected to use this device to improve the design and performance of the UHCPV system, so as to reduce the high temperature factor that is detrimental to the performance of SCs caused by high irradiance. Shi et al. [80] investigated the effect of temperature on the electrical conductivity of polysulfide electrolytes and the optoelectronic properties of QDSSCs. The experimental results found that the increase of temperature was beneficial to the increase of electrolyte conductivity and the transport of polysulfide ions in the electrolyte. However, the increase of temperature caused the increase of dark reaction and the exfoliation of QDs, so the performance of the SC decreased with the increase in temperature.

The efficiency of SC is the ratio of output power to the amount of solar energy irradiated on the surface. The performance of the PV system depends on the output power of the components, which is related to the cell characteristics and environmental conditions [81]. As the light intensity

Fig. 6 Model elements of the proposed ultra-high concentration setting [79]



continues to increase, the ambient temperature around the SCs and the working temperature inside the SCs will therefore continue to rise. This phenomenon will seriously affect the performance of the SC and ultimately affect the efficiency of the SC. For a specific PV system, the output power mainly depends on the operating temperature of SCs and solar irradiance [82–86]. The efficiency of the PV system when operating under actual outdoor operating conditions will be reduced compared with the environment in the laboratory. This is mainly due to the sensitivity of the SC to environmental factors. Climatic parameters have a great influence on most PV technologies which will cause the performance and service life of SCs to be significantly reduced [87]. Xu et al. [88] used a self-built experimental platform (as shown in Fig. 7) to study the influence of dust deposition and environmental wind speed on the temperature of polluted PV glass panels. The research results showed that the deposition of lime soil would cause the temperature of the PV panel to rise, which led to an increase in the temperature of the SCs and a decrease in efficiency. And the temperature of the PV panel decreased with the increased of wind speed.

Considering that the convective heat transfer between wind and PV panels can cause fluctuations in SCs temperature and performance, Hu et al. [89] established a new model for the convective heat transfer coefficient with dust-free deposition. The results showed that the convective heat transfer coefficient of PV panels first increases and then decreases with the increase of dust accumulation density. And the average heat transfer coefficient of dusty PV modules is slightly higher than that of clean PV panels by 4.13%. On the one hand, the increase of convective heat transfer coefficient can reduce the surface temperature of SCs, but on the other hand,

the loss of convective heat transfer will reduce the absorption of solar radiation by SCs. This point needs further comprehensive consideration. Tiano et al. [90] developed a model capable of estimating the temperature effect of PV panels mounted on automobiles under real meteorological conditions. Through model testing, it was found that the increase in the temperature of the PV panel during the parking phase resulted in a significant decrease in its efficiency. Khanna et al. [91] compared the performance analysis of PSCs, DSSCs, and m-Si SCs under different operating conditions. The results showed that as the wind direction angle increases from 0° to 90° , the temperature of all three cells increases, and the cell efficiency correspondingly decreases. When the wind speed increased from 0.5 m/s to 6 m/s, the temperature decreased and the efficiency increased. As the panel inclination increased from 0° to 90° , the temperature decreased and the efficiency increased. And as the solar radiation increased from 200 W/m^2 to 1000 W/m^2 , the cell temperature increased and the efficiency increased. But m-Si SCs generate higher heat resulting in higher temperatures than PSCs and DSSCs. Jošt et al. [92] evaluated the performance of PSCs in field tests on rooftops exposed to real weather conditions. The test results indicated that PSCs exhibited excellent stability at temperatures up to 358.15 K. The performance degradation at high temperature is closely related to the drop in V_{oc} . Chen et al. [93] investigated the long-term stability of DSSCs under alternating temperature (233.15–358.15 K), extreme high temperature (358.15 K), and extreme low temperature (233.15 K) conditions. The research results showed that extreme low temperature had no obvious effect on the stability of DSC, and the stability of DSC was mainly determined by the high-temperature stage. The long-term high-temperature test would not only affect the sealing of the SCs, but also adversely affected the interface of the SCs, reduced the electron transport life of the SCs. Pan et al. [94] studied the effect of partial shading on the characteristics of DSSC modules. The study found that the temperature rise of DSC modules caused by partial shading was much slower and smaller than that of silicon SCs, and the characteristics of monolithic DSCs under short-term shading remain stable. For DSC modules operating in charging mode, the maximum power point (MPP) and operating point would change when shadows appear.

3.2 Non-uniform temperature distribution

Due to the influence of external conditions such as sunlight and wind conditions and internal factors such as SC loss and heat production, non-uniform temperature distribution is common in SC applications. Which has a great impact on the performance of SCs. The uneven temperature distribution of SCs and current mismatches and hot spots on the cells cause the efficiency of SCs to decrease or thermal stress will cause

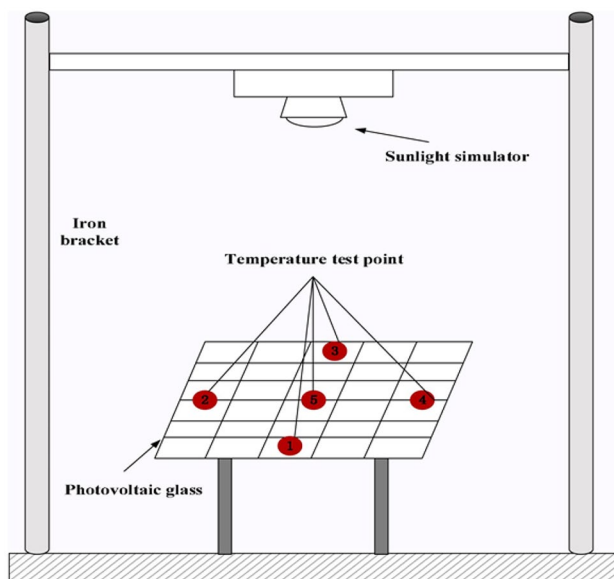


Fig. 7 Schematic diagram of experiment building platform [88]

permanent structural damage [95]. The damage to SCs will further affect the PV system. And the uneven temperature distribution will affect the performance of PV system in two ways: (i) due to the loss of output power, the system has experience efficiency loss; (ii) temperature changes cause thermal fatigue to cause irreversible damage, and excessive local heating reduces the reliability of the system [96].

Researchers have conducted detailed studies on the reasons for the non-uniform temperature distribution of SCs and the impact of this effect. Lu et al. [97] used the two-dimensional finite difference method to analyze the influence of ambient temperature and the temperature difference between the front and back of the SCs on the output parameters of a typical commercial c-Si SC. The results showed that the deviation of the internal temperature distribution of the cell from the ideal temperature distribution was mainly caused by three thermal mechanisms: Joule heat, heat absorbed or released in the process of carrier generation and recombination, and heat generated by the Peltier and Thomson effects. Aiming at the exploration of the formation process of the non-uniform temperature distribution of the SCs, Zhai et al. [98] discussed the temperature rise behavior during the photoelectric conversion process and the change of the internal temperature of the SC under the non-uniform working condition. The study found that under the same light conditions, the area with a higher initial temperature of the cell absorbed more heat than the area with a lower initial temperature, and the final temperature was higher. The temperature rise difference in the high and low temperature areas was approximately proportional to the initial temperature rise difference. In addition, the deterioration effect caused by uneven temperature distribution was also observed, which would reduce the PV performance of the SC and make the PV system lose its optimal operating state. In the application of SCs, due to some external conditions such as human operation or external shading, one part of the SC will be exposed to excessive light, and the other part will be exposed to very little light, causing uneven illumination, which will also lead to the temperature distribution of the SC is uneven. Nasby and Sanderson [99] showed that due to the characteristics of optical illumination, the temperature distribution of SCs is uneven and their efficiency is reduced. The uneven temperature distribution of SCs is mainly caused by SC degradation and excessive wear caused by uneven illumination [100]. Since then, many researchers have done research on this issue. Coventry et al. [101] analyzed the temperature change of a single PV system. The internal temperature of the cell showed that there was a temperature difference of up to 287.15 K between the middle and the edge of the cell. The uneven illumination strongly affects the temperature distribution on the SC. Li et al. [100] summarized the impact of uneven illumination and the resulting uneven temperature distribution on SC in a review literature.

They showed that the uneven temperature distribution of SCs can also cause excessive temperature, which is harmful to the SC and the entire system. One can see that the uneven distribution of incident light will have an important impact on the internal temperature of the cell and thus affect the efficiency of the cell.

Recently, the CPV system has attracted more and more attention from researchers due to its advantages in reducing costs while maintaining high efficiency. In the CPV system, a concentrator is used to concentrate a large area of light on the SC, which improves the energy density of solar radiation. However, this also leads to uneven temperature distribution of the SC, which will reduce the efficiency of the SC to a certain extent. Baig et al. [102] reviewed the causes and effects of inhomogeneity in CPV systems, and emphasized the importance of this issue in SC design. And they reviewed the SC characterization methods under non-uniform flux conditions. Finally, they proposed a method to reduce the unevenness of the concentrating SC to improve the performance of the concentrating SC. To limit this reduction in efficiency, Gupta and Langelaar [103] proposed a new method of optimizing the metallization design of SCs-topology optimization to adapt to certain uneven lighting and temperature distributions. The metallization design obtained through topology optimization significantly improves the power output of the CPV cell. Aiming at the problem that the uneven distribution of solar radiation absorbed by the concentrator will have an important impact on the CPV system, He et al. [104] established a multi-scale-multi-physics integrated mathematical model. And they applied it to the physical model of the CPV system. Studies have found that the high temperature difference caused by the low reflectivity nanostructure was one of the important reasons for the difference in the non-uniformity and uniformity of the photoelectric characteristics. In addition, different wind conditions will also have an important impact on the temperature distribution of SCs [105]. Dabaghzadeh and Eslami [106] used the finite element method to numerically solve the control momentum and energy equations to study the temperature distribution of SCs under different wind conditions. The simulation results showed that the wind field plays an important role in the temperature distribution and performance of SCs. To study the effect of wind conditions on SCs in detail, Singh et al. [107] studied the effects of wind direction, wind speed, inclination and solar irradiance on SC temperature distribution, heat loss and cell efficiency. The results showed that the cell temperature increases with the increase of the azimuth angle of the wind, and decreases with the increase of the wind speed. And the temperature in the middle part of the cell is the highest, and the surrounding temperature is relatively low.

In addition, the potential for hidden cracks in the cells inside the modules and other potential quality issues can also

cause uneven operating temperature distribution when used for a period of time. In summary, the uneven temperature distribution of the SC is caused by a variety of reasons, and it is difficult to completely avoid this problem in the actual application of the SC. By analyzing the mechanism, we hope to shed light on future research.

3.3 Types of SCs and its functional layers

Understanding how SCs react with temperature and how their performance is affected by these changes is critical for creating suitable and reliable devices [36]. According to reports, more than 90% of the global PV market relies on c-Si based SCs [108]. An in-depth understanding of the temperature-dependent properties of silicon SCs is critical [109]. Roshi et al. [110] designed and simulated a silicon-based SC using PC1D numerical simulation software, and the schematic diagram of the silicon-based SC is shown in Fig. 8. It was observed that I_{sc} remained constant while P_{max} and V_{oc} decreased when the temperature increased. And they calculated the optimal value of temperature based on the I-V and P-V curves of the silicon-based SC. At 298.15 K, the conversion efficiency was 22.58% for 150 μm p-type layer thickness. However, when using PC1D to optimize the p-type layer thickness to 200 μm , the efficiency increases to 24.25% at 273.15 K. Makhlof and Shehata [111] successfully prepared a complete (Al/Ag/MoO_x/Ag/MoO_x/n-Si/MgO/Mg/Al heterojunction SCs (OMO-Si HJSCs)) by conventional thermal evaporation technique. The engineering structure of the SC was fabricated as presented in Fig. 9. It had an efficiency of 8.80%, the V_{oc} of 554 mV, a J_{sc} of 32.25 mA/cm², and the FF of 49.25%. According to I_s and C-V measurements, the device's performance degrades with increasing temperature, with decreased recombination resistance, lifetime, depletion width, and built-in potential.

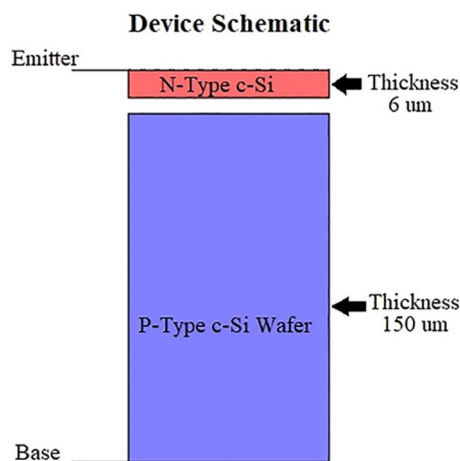


Fig. 8 A schematic of silicon SC model [110]

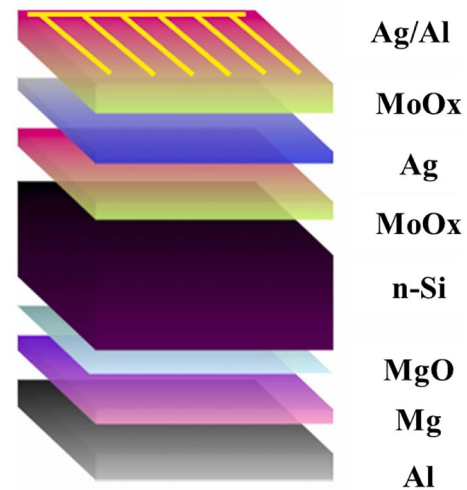


Fig. 9 The engineering structure of the Ag/Al/MoO_x/Ag/MoO_x/n-Si/MgO/Mg/Al heterojunction SC was fabricated as presented [111]

The OMO-Si HJSCs have the highest lifetime (957 μs) at 298 K and the lowest lifetime (5.36 μs) at 433 K. Li et al. [112] measured the effect of temperature on the PV performance of an aluminum alloy back-junction n-type silicon SC. Due to the unique spectral response, stronger dependence on carrier diffusion length, and incomplete ionization of aluminum, back-junction silicon SCs have a lower absolute temperature coefficient compared to conventional front-junction SCs, especially for medium carrier diffusion lengths. The temperature coefficient is lowest when compensation silicon is used as the base layer. Such SCs can generate more electricity at high temperatures. Chatterji et al. [30] explored the functional dependence of temperature coefficient on carrier selectivity silicon SC material and interface properties. It was found that when the interface quality is not perfect, the temperature coefficient increases with the band discontinuity. For larger band discontinuities, the temperature coefficient increases with temperature independent of the interface quality. The results showed that the passivation quality of the minority carrier layer interface is more critical than that of the majority layer interface. And the temperature coefficient of a trap structure with traps at both interfaces is better than a single device structure with traps at only one interface (ESL/Si or HSL/Si). This means that the temperature coefficients do not accumulate.

The thin-film SCs reduce the use of silicon materials, opening up the use of semiconductor materials in SCs [3]. Typical thin-film SC materials include α -Si, CIGS, and CdTe [30]. Ahmed et al. [34] reported the effect of individual functional layers within a Cu(In,Ga)(S,Se)₂ SC stack on the temperature dependence of the overall cell performance. The experimental results showed that the temperature dependence of the power efficiency is most affected

by the absorber layer. And the temperature dependence of power efficiency was mainly controlled by the change of V_{oc} . Bagiński and Gupta [37] measured the temperature dependence of bulk heterojunction OSCs fabricated from poly(3-hexylthiophene) (P3HT) and [6,6]-phenyl-C61-butyric acidmethyl ester (PCBM) of light absorption, PL, quantum efficiency (QE), I-V parameters, and overall efficiency in the range of 413.15 to 263.15 K. The measurements found that the absorption of both the P3HT structure and the P3HT/PCBM blend decreased with temperature, and the absorption spectra showed a red shift, increased peak absorption, and a sharper shoulder. The absorption of the PCBM structure did not vary with temperature. As the device was annealed, the QE of the device increased dramatically with increasing temperature. The I-V characteristics of unannealed devices increased with annealing and changed with temperature after annealing. The efficiency peak appeared at an average temperature of 299.15 K and began to decay. Dong et al. [113] first reported a high temperature (over 473.15 K) PSC. The device maintained more than 80% of its initial PCE after heating at 473.15 K for 45 h. Compared with conventional high-temperature SCs (Si, CuInGaSe, and GaAs), the CsPbI₂Br device exhibits excellent V_{oc} and FF temperature coefficients, but it has a poor J_{sc} temperature coefficient.

In response to the problem of solving the reachable temperature of the SC active layer, Gallardo et al. [114] measured the temperature of TiO₂ films in DSSCs applying the classical method of Raman spectroscopy using the Stokes peak and the inverse Stokes peak. The effect of rising temperature. Measurements showed that the properties of semiconductors are directly dependent on temperature. And a larger temperature increase was observed in thicker films. Kou et al. [115] investigated the intensity dependence of light intensity for a thin titania barrier layer on transparent conductive oxide (TCO) in DSSCs. It was found that electron recombination through TCO became more and more evident with increasing temperature. The resistive layer on the TCO surface remains fairly stable without degradation over a wide temperature range, while the efficiency of the device without the barrier layer can drop by more than 10% at high temperature. Fernandes et al. [39] investigated the effect of substrate temperature on the charge transport properties of Ta₂O₅ cathode buffer layers in inverted polymer SCs. The increase in defect states deposited at 523.15 K was found to be benign by PL measurements. As the substrate temperature was increased to 523.15 K, the PCE increased from 2.8% to 3.1% by current density–voltage measurements, a ratio/percentage improvement of 10.7%.

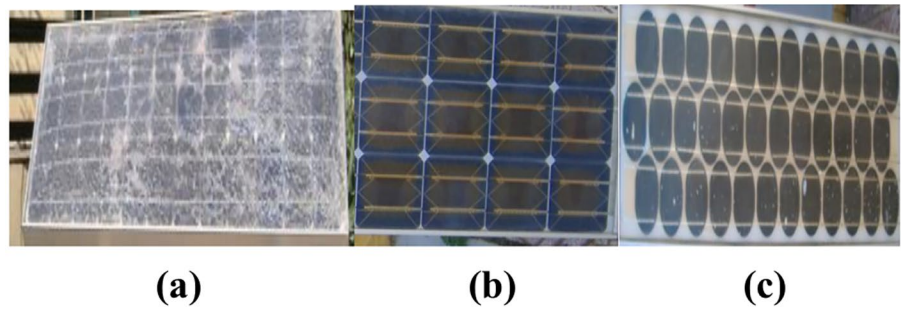
3.4 The manufacturing process

High-temperature sintering, soaking, and annealing are indispensable links in the preparation process of SCs. Lee

et al. [40] investigated the effect of processing temperature on diketopyrrolopyrrole-alt-thieno[2,3-b] thiophene polymers for high-mobility thin film transistors and high open circuit voltage polymer SCs. In the OTFT device, it had a high hole mobility of 0.16 after heat treatment at 423.15 K. The pDPPTTi-OD:PC71BM blend film treated at 363.15 K showed a PCE of 2.9%. The results showed that high temperature treatment helps to control the morphology of semiconducting polymer or polymer blend films. Annealing is a necessary step for film formation [70]. Hussain et al. [116] investigated the effect of post-annealing on the structural, optical, and electrical properties of CuSbS₂ thin films prepared by combined thermal evaporation techniques. The calculated optical band gap after annealing at 623.15 K was found to be 1.58 eV, which is close to the optimal value for high-efficiency SC materials. Meanwhile, it was also observed that the resistivity of the samples decreased with increasing annealing temperature (12.1–1.43 KΩ-cm). Vanalakar et al. [117] reported a pulsed laser deposition route to synthesize compact single-phase Cu₂SnS₃ (CTS) films and investigated the effect of annealing temperature on the formation of CTS films in ternary chalcogenide semiconductor CTS SCs. The research results showed that the annealed CTS films have polycrystalline properties of cubic crystal structure, and the crystal size of the CTS films increases with the increase of the annealing temperature. The direct optical bandgap energy of the CTS film annealed at 673.15 K is 1.01 eV and reaches the optimal crystal size.

Paramitha et al. [118] investigated the effect of sintering and soaking temperature on the performance of DSSCs. The TiO₂ films were sintered at different temperatures from 473.15 to 773.15 K for 60 min. From the efficiency test, it was found that the best performance of DSSC occurred when the TiO₂ film was sintered at 723.15 K. In addition, the dye soaking temperature was evaluated from 313.15 to 323.15 K for 6 h. The optimal immersion temperature of DSSC is 323.15 K, and the maximum efficiency is 2.621%. Buldu et al. [119] investigated the effect of sulfurization temperature on the morphology, composition, and structure of Cu₂ZnSnS₄ thin films grown on titanium (Ti) substrates. The results of the study show that at a sulfidation temperature of 843.15 K, the grain size is optimal and the copper-poor/zinc-rich composition required for SC applications is also obtained. Operating temperature also plays a crucial role in SCs. Ouslimane et al. [120] reported the effect of operating temperature (300 to 440 K) on the performance of MAPbI₃ SCs based on ZnO electron transport materials. The results showed that the SC achieves the highest efficiency at a low temperature of 300 K. The increase in temperature affects the mobility of holes and electrons as well as the carrier concentration, resulting in a decrease in the efficiency of PSCs. In addition, J_{sc} increases slightly with increasing temperature due to the reduction of the energy band gap and

Fig. 10 Degradation mode of PV module UDTS (50) installed in warm and humid [122] **a** Broken glass **b** Discolouration cells **c** Yellowing EVA



the generation of more electron–hole pairs. The decrease of V_{oc} with the increase of operating temperature will lead to more interfacial defects and increase of series resistance and decrease of carrier diffusion length.

Macroscopic temperature effects such as light intensity and uneven temperature distribution directly affect the operating temperature of SCs. The electrical parameters of SCs, except J_{sc} , all decrease with the increase of temperature, which also includes the efficiency of SCs. Changes in electrical parameters will also directly affect optical parameters such as carriers and band gaps inside the SCs, as well as internal defects.

4 Other factors affecting the performance of SCs

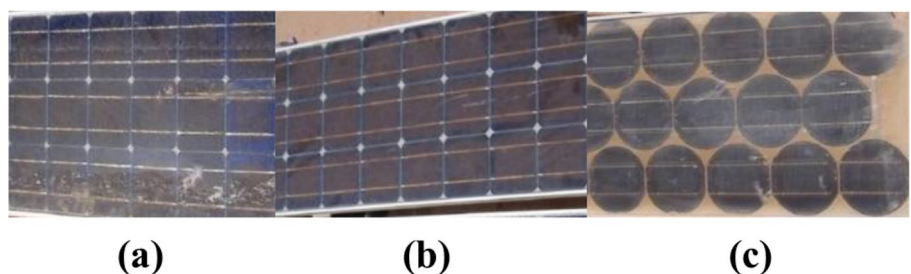
As seen in the previous sections, this temperature effect severely affects the electrical performance parameters of the SCs. This negative impact on the SCs will eventually reduce the efficiency of the whole PV system. In addition to the temperature effect on SCs discussed in detail above, the efficiency of SCs is affected by many other factors. This section will conduct a comprehensive analysis of other factors affecting SCs, so that we have a more comprehensive understanding of the PV system, and do some theoretical paving for further optimization of its performance.

4.1 External factors

After the influence of long-term solar radiation and external environment, the aging phenomenon of PV panels will

occur. The degradation of PV modules mainly depends on the influence of four factors: temperature, irradiation, humidity, and mechanical stress. The influence of these factors will further cause the degradation of SC, such as corrosion, discoloration, delamination, and damage [121]. The previous only described the influence of the external environment on the temperature effect of SCs, and the effect of aging on the performance of SCs has not been clarified. Bandou et al. [122] simulated UDTS (50) to test the aging phenomenon of PV modules and analyzed the performance of PV modules. It can be seen from Figs. 10 and 11 that under the action of environmental factors, the PV module had undergone glass breakage, discoloration, and yellowing of the cell. And the discolored and yellowed surface of the cell was caused by long-term exposure to ultraviolet radiation and high temperature. The effect of these environmental factors will eventually lead to the aging of the cell, which will reduce the power of the cell. In addition, PV efficiency and cell capacity degradation vary with time and cycle number, respectively [123]. The existence of defects such as light-induced degradation (LID), hot spots, corrosion, or delamination will affect SCs and the constituent materials used to protect them, and ultimately lead to cell aging. Azizi et al. [124] studied the impact of PV module aging on production from the perspective of harmonics and power reduction. And they considered a hybrid scheme of instantaneous maximum power point tracking (MPPT) and long-term degradation. In this study, it was found that the MPP drops at a rate of 1% per year. The resistance change was estimated to be 12.8%, and the receiver power affects nearly 30%. Because cell life was highly dependent on operating strategies and aggressive charge/discharge operations,

Fig. 11 Degradation mode of PV module UDTS (50) installed in Adrar (warm and dry) [122] **a** Broken glass **b** Discolouration cells **c** Yellowing EVA



cell capacity ages faster, and overall operating costs were higher due to more frequent replacement.

Furthermore, due to different PV materials having different absorption characteristics, their roles in SCs are also different. As mentioned earlier, the performance of SCs can be affected by the temperature effects of different functional layers. Wang et al. [125] studied the η of DSSCs based on 1 mol% Sb-doped TiO_2 electrode to reach 8.13%, which was significantly higher than that of undoped DSSCs. Wu et al. [126] reviewed the cell performance of the third-generation PV technology using nanomaterials. And they showed that nanomaterials and nanostructures severely affect the optical and electronic properties of the functional layer of SCs by adjusting the morphology, microstructure, and surface state of the functional layer. This phenomenon significantly affected the output voltage and conversion efficiency of SCs. Ajayan et al. [127] reviewed SC based on different materials. In this study, it was concluded that the highest conversion efficiency of silicon, PVK, DSSC, III-V SC, flexible SC, QDSCs, and OSCs were 27.6%, 25.2%, 12.3%, 53.8%, 15.38%, 18.05%, and 13.76%. Among them, the PC of MJSC based on III-V group materials was more than 50%. However, it would be affected by carrier, light absorption, and thermal energy loss. The operational stability, reliability, and large-area device efficiency of PSC still need to be further improved. And the PVK/silicon hybrid series SC is effective in improving PCE. The poor carrier mobility of OSCs limits their PCE, but the series cell technology can effectively improve the PCE of OSCs. Li et al. [128] successfully fabricated novel composite oxide TiO_2 - SrTiO_3 thin films on FTO substrates of SC. The test results showed that the film not only has high sensitivity, but also has relatively stable photoelectric activity.

In addition, the series–parallel connection of external circuits also has a certain impact on the efficiency of SCs. Yusof and Baharuddin [129] studied the output current of SCs in a series or parallel arrangement within a certain range of voltage, current, and power in a solar system. Their results showed that the series output current value was higher than

the parallel output current value. And the series and parallel output voltage values were the same. But the parallel arrangement was more efficient than the series connection.

4.2 Internal factors

According to reports, the charge trap state plays an important role in the degradation of PSC and other devices [130, 131]. Understanding the distribution of trap states in space and energy is one of the most basic elements to understand the impact of charge traps on charge transport and recombination in PVK materials and devices [132]. Shintaku et al. [133] used the temperature dependence of V_{OC} to quantify the V_{OC} loss associated with the energy of the charge transfer state, and studied the effect of trap-assisted recombination on the open circuit voltage loss of phthalocyanine/fullerene SCs. The research results showed that as the temperature increases, the order of charge recombination decreased from 5.35 to 3.05, which indicated that the trapped charges were being released and became free charges. And the trap state played a key role in charge recombination and V_{OC} loss. To reduce the effect of defects, Chen et al. [134] selected the natural amino acid molecule methionine as an effective passivator to enhance the optoelectronic properties of MAPbI_3 thin films.

Many researchers have pointed out that point defects and grain boundary passivation also play a key role in achieving high-efficiency PSCs [135, 136]. These defect sites would not only hinder high photovoltage, but also shorten the life of PSC [137–139]. Some researchers pointed out that by optimizing the alkyl chain length as a grain and interface modifier, which can promote favorable grain orientation and suppress the density of trap states to improve photoelectric properties [140]. Yuan et al. [141] used two kinds of amino acids, cyclic proline and linear glycine to study the steric effect of Lewis acid–base as PSC additives. The growth process of two kinds of amino acids with different grains is shown in Fig. 12. PSC with PVK added with proline as an additive showed the highest

Fig. 12 Schematic diagram of the growth process of two kinds of amino acids with different grains [141]

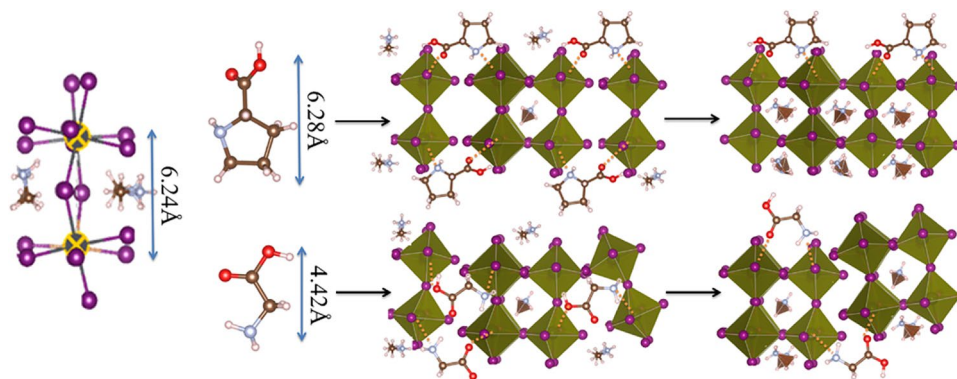
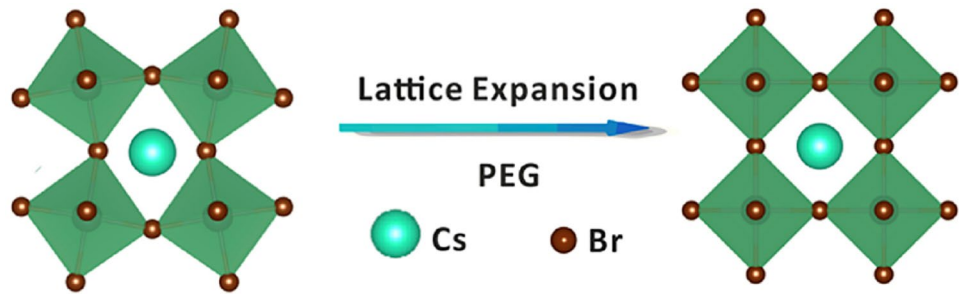


Fig. 13 Schematic diagram of crystal structure change due to local deformation and lattice expansion [142]



efficiency of 19.6% under 1 daylight AM1.5. The PSC with glycine additive was 18.5%, and the original PSC was 17.8%. Proline inactivated the surface and GB trap states in PVK, greatly reducing the charge recombination in PSC. Therefore, Lewis acids and bases with appropriate spatial configuration and size obtained the best performance in PVK grain growth and trap state passivation. Ren et al. [142] added polyethylene glycol (PEG) polymer material before the high-purity CsPbBr_3 PVK film solution prepared by a one-step spin coating method to study its performance on CsPbBr_3 devices. The research results showed that, as shown in Fig. 13, the addition of PEG polymer material would cause the uniform lattice expansion in the CsPbBr_3 thin layer and reduce the trap density. The lattice expansion caused by the polymer is beneficial to improve the performance of the CsPbBr_3 device, thereby improving the PCE.

It can be seen that there are many factors that affect the performance of SCs. From the outside to the inside, from the micro to the macro, from the molecular structure to the light intensity, only by understanding the various factors that affect the performance of SCs, can we better design and improve SCs, so as to produce higher efficiency, so that solar energy can be better utilized.

5 Countermeasures and plans

The efficiency of PV systems with cooling technology is reported to be 52% higher than those without cooling technology [143]. It can be seen that cooling technology is crucial for the conversion efficiency of PV panels. And the cooling technology can also extend the life of PV panels [144]. In the design and application of SCs, the use of suitable cooling technology for different SCs is a factor that needs to be considered. At present, the measures to deal with the temperature effect of SCs generally include those shown in Fig. 14.

5.1 Adding cooling systems to PV systems

Currently, cooling systems are available in two main forms: active cooling and passive cooling. The most fundamental difference between active cooling and passive cooling is that passive cooling does not require energy input [145]. Currently, among passive cooling technologies, PCM cooling has a large potential [146–152]. And the most common active cooling technology is based on thermal conduction and convection heat transfer of the flowing working fluid (e.g., air, water, nanofluids (NFs)). Air has the advantages of small volume change and environmental friendliness under

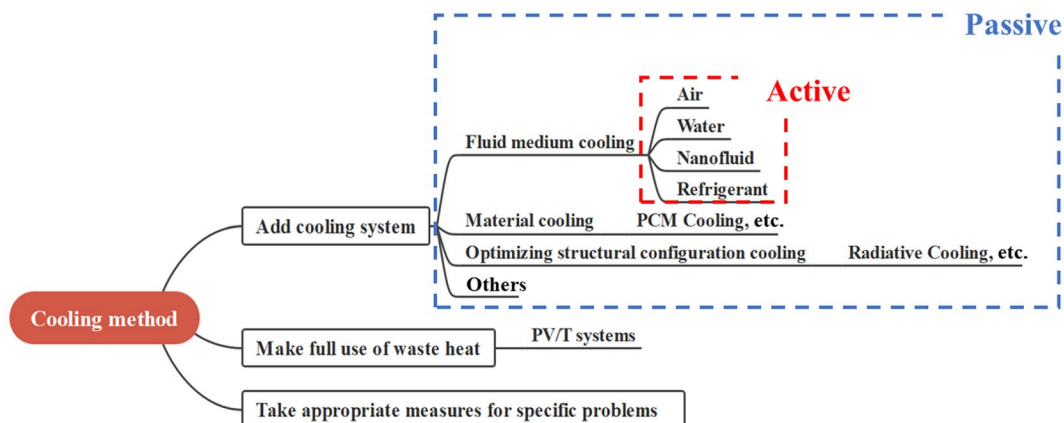


Fig. 14 Classification of cooling methods for PV modules

Table 3 Several typical cooling technologies

Cooling technology	Purpose	Content	Result	Ref
Fluid medium cooling	Active	• Lower panel temperature • Improve electrical efficiency	• Using PCMs leads to a significant increase in natural and forced convection situations [173]	[173]
		• Improve the performance of air-cooled PV/T system	• The reduction in air channel depth improves thermal efficiency but has no significant effect on electrical efficiency • With the increase in air mass flow, thermal efficiency increases and the electrical energy generated by PVs increases slightly	
		• Increase performance of SC • Prevent overheating of PV panel systems	• The best cooling capability and performance are obtained using 15 fins with a copper base and fin HS materials • The decrease in temperature is 283.35 K • The increase in efficiency is 2.74%	
Water	Passive	• Reducing the exergy loss of the air cooling system in the PV/T system	• The exergy performance is increased from 20 to 28% while using air cooling duct with longitudinal fins and inclined baffles	[176]
		• Improving the efficiency of PV modules in hot climates regions	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate • Reduce the average surface temperature on the back of the PV panel from 317.95 K to 303.45 K • Average surface temperature on the front side decreased from 319.75 K to 310.05 K • Electrical efficiency of PV/T modules increased by 8%	
		• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
Water	Active	• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	[177]
		• Improving the efficiency of PV modules in hot climates regions	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
		• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
Water	Active	• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	[178]
		• Improving the efficiency of PV modules in hot climates regions	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
		• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
Water	Active	• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	[179]
		• Improving the efficiency of PV modules in hot climates regions	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	
		• Reducing the exergy loss of the air cooling system in the PV/T system	• The longitudinal fins with inclined baffles provided higher friction factor, thereby ensured higher heat transfer rate	

Table 3 (continued)

Cooling technology	Purpose	Content	Result	Ref
	Passive	• Eliminate the adverse effects of waste heat	• Investigates the performance of a partially submerged floating photovoltaic system (PSFPV)	[180]
	Active	• Reduce temperature effects • Improve PV efficiency • Reduce carbon emissions	• Up to 20.76% more electricity than the land-based PV system • Alleviating the emission of CO₂ • The power and efficiency of SiO₂/water increased by 54.29% and 3.35% • The power and efficiency of Fe₃O₄/water increased by 52.40% and 3.13%	[181]
	Passive	• Reduce temperature effects • Improve hybrid PV thermal system performance	• Five different cooling methods are investigated experimentally, namely natural cooling, forced air cooling, water cooling, SiO₂/water NF cooling, and Fe₃O₄/water NF cooling • Proposed a mechanism to cool PV panels by circulating NFs around them	[182]
	Active + Passive	• Improve PV cooling capacity • Addressing the limitations of PCM temperature degradation	• A new system using a heat transfer medium between PCM and CPV to transfer the dissipated heat from CPV to PCM is proposed	[156]
Materials cooling	NF		• HPVTS show the best performance at NF concentration 0.5	[183]
	Passive	• To obtain lower temperature of dense-array SCs in high concentrating PV system	• Presents an optimized ethanol phase-change immersion cooling method	[183]
	PCMs	• Improve thermal conductivity during phase transition	• The proposed system achieves 60% reduction in the CPV average temperature compared to the conventional direct PCM-PV and water-cooling individual systems • Using nanofluid as HTF enhancer increases the CPV efficiency by 2.7% and reduces the PV maximum temperature and the PCM melting time by 4 °C and 12%, respectively	[184]
	Passive	• Reduce cost and cross the stability and PCM change difficulties • Improve thermal conductivity of PCMs	• The exergy efficiency of the PV/T module with metal foam filled with PCM is 16.7% • All three PCMs reduce the temperature of the PV panel • Hybrid paraffin have better productivity	[185]
		• To improve the performance of the module under high temperature conditions	• Evaluate the impact of simultaneous dual surface cooling on the output performance of PV module experimentally	[144]
			• Panel temperature dropped by 296.7 K • Panel output power increased by about 30.3% • Cooling panel electrical efficiency increased by 11.9%	

Table 3 (continued)

Cooling technology	Purpose	Content	Result	Ref
Optimizing structural configuration cooling	- Eliminate temperature effects - Improve solar energy absorption	Propose a selective plate-based PV/RC hybrid system. It can not only generate electricity through day and night PV conversion, but also obtain cold energy through RC	6.9% higher day/night power output 30.5% increase in night RC power	[186]
	- Eliminate temperature effects - Improve the cooling efficiency of RC	Report a new class of radiative sky cooling material – cellulose-upgraded polymer films	Cools to 277.12 K below ambient when exposed to direct sunlight	[187]
	- Solve the high energy consumption and environmental problems of cooling technology - Improve the cooling efficiency of RC	Propose a new strategy for the PVRC-TE system	- Realize 24-h all day operation - Improve the comprehensive cooling effect and reduce the space heat load - The maximum cooling energy gain in a day of the system is 285.57 MJ/m² - it can be radiatively cooled by 275.15 K	[188]
	- Eliminate temperature effects - Improving the efficiency of flat PV panels	A periodic pyramid-textured polydimethylsiloxane film is presented for RC encapsulating commercial silicon SCs		[189]
	- Eliminate the adverse effects of waste heat - Improving the transmittance of SC cover	A high-performance and scalable radiation cooler for SCs is proposed	- Maintaining an excellent solar transmittance over 0.94 - Under solar radiation of about 850 W/m², the porous cooler can cool the bare silicon wafer by about 7.5 K - Achieve a maximum temperature drop of about 35.6 K - Improve annual output electricity of the SC reaching about 16.5 kW h/(a·m²)	[190]
	Improve the efficiency of SCs	A new cooling technique consisting of a HS and microchannels for cooling SCs is reported	- Decrease the cell temperature, thermal efficiency, and temperature gradient on the cell surface - Increase the cell efficiency, electrical power, and output net power	[191]
	Eliminate the adverse effects of waste heat	Introduce an experimental study for concentrating SC performance cooled by using flat heat pipe	Cooling the cell by heat pipe increases its output power by 24.3% compared to free convection without utilizing heat pipe at incident energy 500 W m⁻²	[144]
	- Reduce the impact of waste heat on CPV systems - Improving the efficiency and reliability of SCs	Exploit graphene as a pre-illumination cooling technique (ND filter) for SC	- Can eliminate the strenuous cooling techniques such as active water cooling and combined heat pipe and sink cooling for CPV - Without cooling aid has shown an excellent cell efficiency	[192]

Table 3 (continued)

Cooling technology	Purpose	Content	Result	Ref
Others	Passive	• Advance the space thermal control technology • Exploit the space solar power • Eliminate the effect of temperature on c-Si SCs • Improve cell stability and efficiency	• Full-spectrum selective thin-film-based photonic cooler is proposed for passively cooling SCs of the space solar power station via orb-shape membrane energy gathering array concept • Photonic structure-based spectrum-selective mirror is designed to cool the operating temperatures of c-Si SCs passively	• Reduce the temperature of SCs by 303.15 K at least • Improving the SC efficiency 1.4% Could make c-Si SCs absorb about 93% of the sunlight in the wavelength range of 0.3 to 1.1 μm and only absorb about 4% of the sunlight in the wavelength range of 1.1 to 2.5 μm

low temperature conditions [153]. However, to get a higher cooling effect, a flow medium with high stability and better thermal conductivity is required. It has been noted that the PV efficiency of NFs cooling based PV systems is higher than those based on water cooling [154]. In recent years, NFs have been widely used in photothermal conversion due to their good stability and thermal conductivity. It is one of the most promising technologies for active cooling [146]. Nasef et al. [155] proposed a new system using NFs between the PCM and the CPV. It eliminated the defects of PCM by transferring the heat emitted from CPV to PCM, which took full advantage of the excellent endothermic properties of NFs. Notably passive radiative cooling (RC) [156, 157] is a recent emerging passive cooling technique. Its principle is to reflect sunlight with a wavelength range of 0.3–2.5 μm and at the same time dissipate its own heat through an atmospheric transparent window with a wavelength of 8–13 μm to outer space. Table 3 lists some recent work on cooling technology through the form of cooling.

5.2 Make full use of waste heat

During the photoelectric conversion process, SCs can only convert part of the solar energy into electricity. The remainder is converted into heat, affecting the performance of the SCs [158]. In view of this, the researchers developed a photovoltaic/thermal (PV/T) system that enables continuous supply through active cooling technology to keep PV module temperatures low. And it can recover part of the waste heat from PV modules to achieve the purpose of cogeneration [159–165]. At present, the PV/T system is very mature, but the temperature effect in SCs cannot be completely eliminated. The negative impact caused by the temperature effect can only be reduced as much as possible. Hachchadi et al. [166] investigated a new solar energy system that couples a PV/T air collector with a water–air heat exchanger. The total area required to heat the system had been reduced by approximately 33%. And the efficiency of PV modules increased by 2.0% in winter and 5.1% in summer. This means an average reduction of 11.4 kg of CO₂ emissions per day. Rejeb et al. [167] developed a new solar PV/T design to improve thermoelectric (TE) performance, as shown in Fig. 15. They found that this advanced channel PV/T design with optical coating can operate at higher fluid temperatures and lower SC temperatures. And it had better electrical and thermal efficiency. Zhang et al. [168] used thermal interface materials in PV/T coupling devices to achieve the effect of improving solar energy utilization. And the experimental coupling device is shown in Fig. 16. The research results showed that coupling the PV system with a thermoelectric generator (TEG) can effectively reduce the temperature rise of the SC and enable waste heat to generate electricity. And coating the thermal interface material

Fig. 15 Design model diagram of a new type of solar PV/T [167]

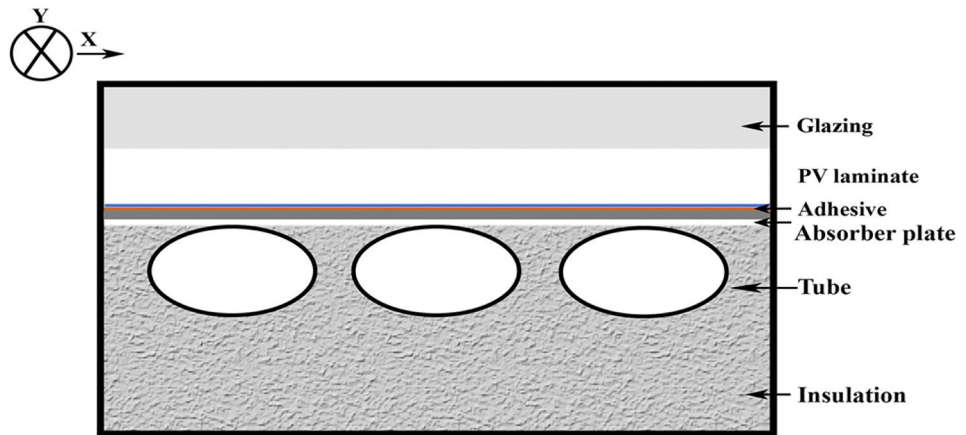
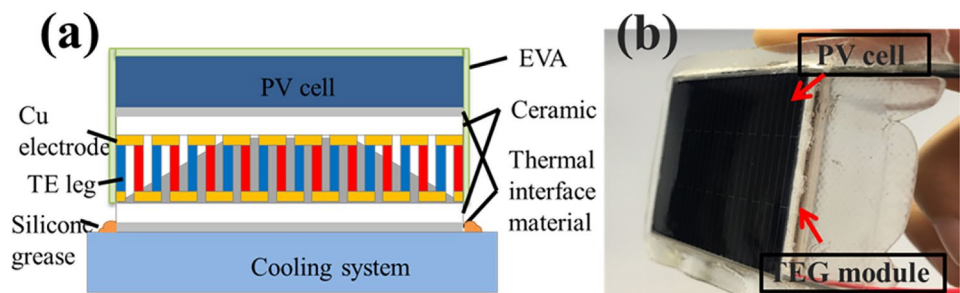


Fig. 16 **a** Structure of PV-TE coupling device **b** Photo of PV-TE coupling [168]



can reduce the contact thermal resistance and enhance the heat transfer. The performance of the PV/TE coupling device was significantly improved. In the same year, the team [169] also manufactured a series of new integrated PV/T hybrid devices with enhanced heat transfer capabilities by removing the upper ceramic plate of traditional TE devices. The research results showed that the PV/T hybrid device manufactured by this method can effectively eliminate the negative influence of the PV system due to the temperature effect, thereby increasing the electrical output of the SC.

5.3 Combination with other technologies for comprehensive energy utilization and PV cell cooling

To solve the problem of rising temperature in the car when parking in the hot summer, Pang et al. [170] designed a DC air conditioning system powered by solar PV modules. The design had a good improvement to the interior environment of the vehicle and was in line with the needs of the human body. Aiming at the important influence of temperature on MJSCs, Djermane et al. [171] conducted a numerical study on the cooling temperature of InGaP/InGaAs/Ge MJSCs under concentrated illumination. The aim is to find the limiting temperature of each sub-cell to introduce a cooling system and avoid degradation of the tandem SCs under

concentrated lighting. Li et al. [172] used their fabricated diffractive microlens arrays for optical micro-ground structures on glass substrates of solar panel devices to create a long-term stable PV system. The results showed that the diffractive microlens array not only reduces the visible light reflectivity by 22.2%, but also increases the infrared light reflectivity from 16.73% to 22.86%. And the average power generation was also greatly improved and lowered the surface temperature of the solar panels by 281.15–283.15 K.

Adopting necessary cooling methods and solutions will greatly reduce the temperature impact on PV modules and improve their performance. The current cooling technology is relatively mature, but continuous improvement and technical breakthroughs in various methods are still needed. A combination of cooling techniques can be considered in future research to reduce the temperature effect of SCs with the maximum, lowest cost and lowest power consumption to improve the thermal stability and efficiency of SCs.

6 Conclusions

In summary, there are various factors that affect the temperature of SCs. The temperature effect of the SC will affect the intrinsic properties of the cell material and ultimately affect its power generation efficiency. This article

reviews the temperature effect of SCs, showing its mechanism and the latest research progress. In addition, other factors that affect the performance of SCs are discussed, and various methods and solutions to deal with the temperature effects of SCs are introduced in detail. The following conclusions can be drawn from this article:

1. The increase in the temperature of the SCs not only reduces the efficiency of the SCs, but also damages the PV cell and reduces their life. Before the application of SCs is truly popular, the challenge of long-term stability must be solved.
2. From a microscopic point of view, as the temperature of the PV cell increases, the increase in the internal carrier recombination rate due to the increase in the carrier concentration will cause the efficiency to decrease.
3. The parameters V_{oc} , I_{sc} , FF, η , etc. that characterize the performance of the PV cell all change with the change of the PV cell temperature. As the temperature increases, the V_{oc} decreases to reduce the cell efficiency, and the I_{sc} increases slightly to increase the cell efficiency. Therefore, the efficiency of the PV cell depends on the compromise between the decrease in V_{oc} and the increase in I_{sc} .
4. Due to the influence of various external and internal factors, the uneven temperature distribution of the PV cell will cause the efficiency of the SC to decrease and damage the PV system.
5. Cooling technology for the temperature effect of SCs is very mature. However, how to minimize the influence of temperature effect on SC and make full use of solar energy is still a hotspot of current research. NF and PCM still have great potential for future cooling technology.
6. In addition, some other important factors that affect the performance of SCs include trap states, spatial effects between internal molecular structures, external environment, SC aging, and materials used. In summary, the temperature effect of SC is caused by a variety of reasons, and it is difficult to completely avoid this problem in the practical application of SC. By analyzing the theoretical mechanism, we hope to shed light on future research.

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Declarations

Competing interests The authors declare no competing interests.

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